

Spingarn's Method of Partial Inverses

Given: $A : \mathcal{H} \rightarrow 2^{\mathcal{H}}$ maximally monotone, closed subspace V

Find: $x \in V, u \in V^{\perp}$ such that $u \in Ax$

Step 1: Initialize $x_0 \in V, u_0 \in V^{\perp}$

Step 2: Compute $y_n = J_A(x_n + u_n)$

Step 3: Compute $v_n = x_n + u_n - y_n$

Step 4: Project $(x_{n+1}, u_{n+1}) = (P_V y_n, P_{V^{\perp}} v_n)$

Convergence : $(x_n, u_n) \rightarrow (x, u)$ **where** $u \in Ax$

Solution to the constrained problem: $x \in V$ and $u \in V^{\perp}$

Key Insight :

Decomposes constrained problem into independent subproblems